

Raw phi1 voltage calibration

RUN

194917_raw-phi1_readback-check

Status: completed; 18 individual PAX readings

MEASUREMENT

0.0 to 0.1 RP command V, step 0.05 V; 1 forward/reverse cycles.

Settling: 1.0 s; 3 consecutive readings per held level.

VOLTAGE GRID

Uniform grid specified above.

PHYSICAL SETUP

Physical position awaiting confirmation; no assumed port

IN1/IN2 physical mapping awaiting confirmation

WHAT IS RECORDED

Requested RP voltage, ASG register readback, full FPGA OUT1 scope trace, explicitly selected IN1/IN2 scope traces, raw PAX records and acquisition times.

WHAT IS NOT ESTABLISHED

Internal OUT1 data is the signal before the DAC. It is not an independent BNC or piezo-terminal voltage measurement. No historical driver gain is applied.

ANALYSIS POLICY

No curve fits, interpolated V_{π} , drift subtraction, or optimized coordinate rotation. Phase is $\text{atan2}(S3, -S2)$; unwrapping only adds 2π at coordinate wraps. Step medians are labelled summaries; all individual measurements remain in data.csv and pax_raw.jsonl.

DATA VALIDITY NOTE

CSV analog_in1_v is the selected scope input (out1), NOT physical ADC IN1. Treat this field as invalid for electrical-input measurement.

Raw phi1 voltage calibration

DATA VALIDITY NOTE

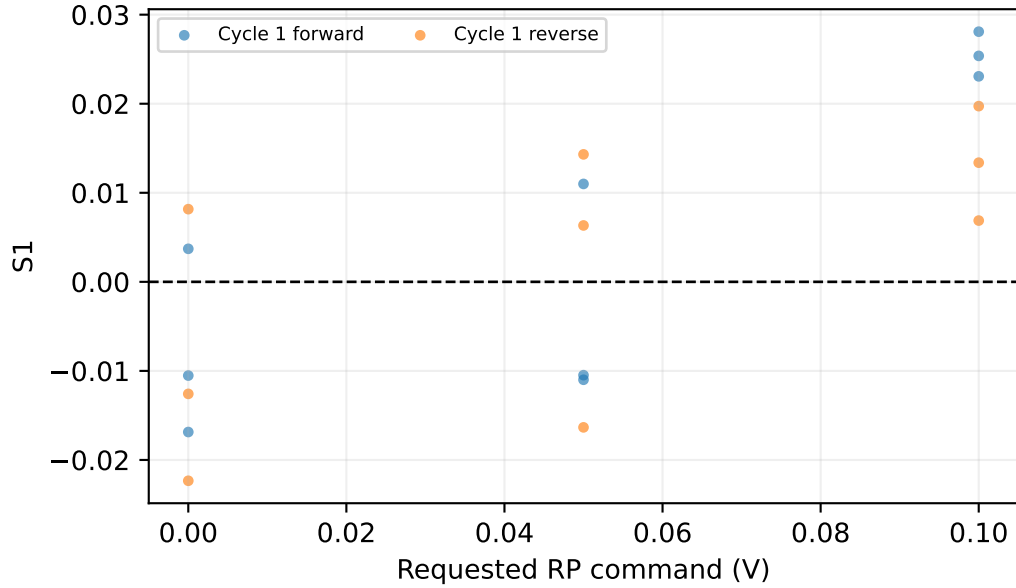
Authoritative physical input waveforms are saved in scope/*.npz: analog_in1_samples explicitly selected in1; samples[1] selected in2. Digital OUT1 is samples[0]. Those traces are valid.

DATA VALIDITY NOTE

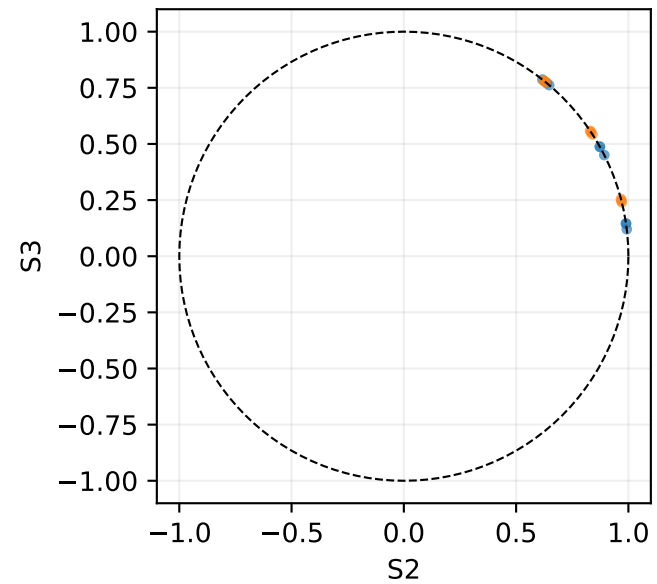
Corrected subsequent runs use explicitly routed saved traces for all scope signal summaries.

Individual PAX readings at held voltages

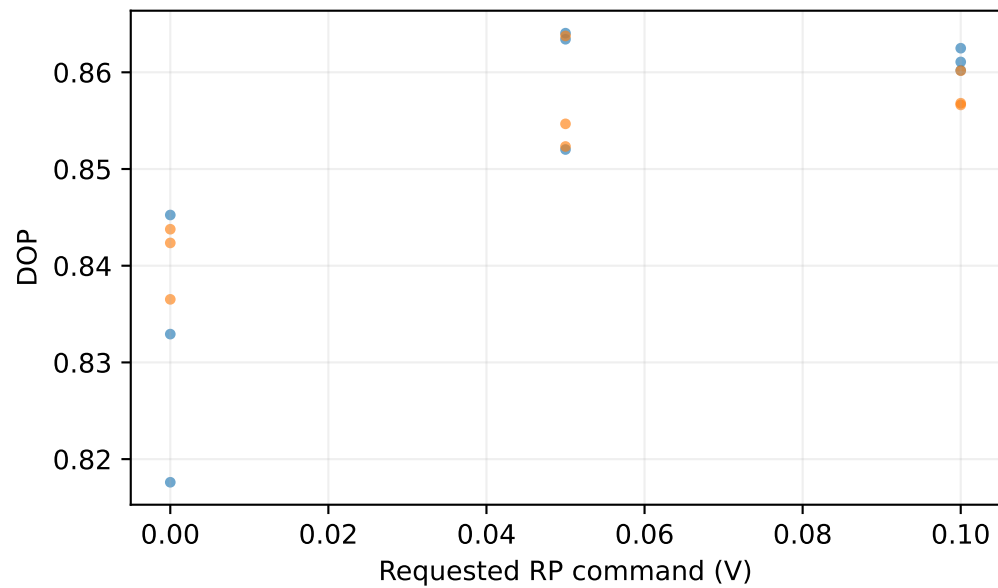
Departure from the $S_1 = 0$ plane



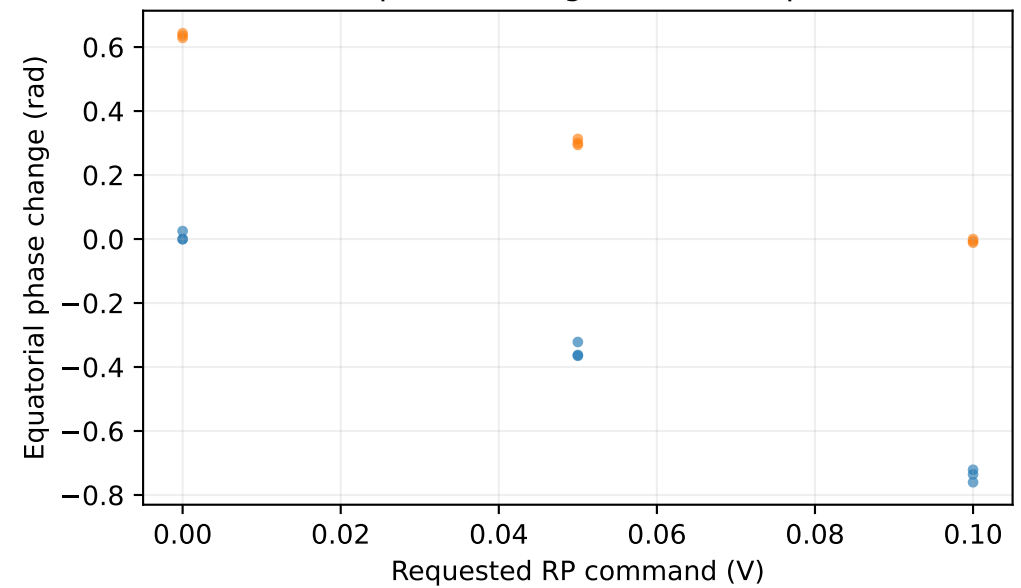
Raw orientation trajectory



Raw PAX DOP



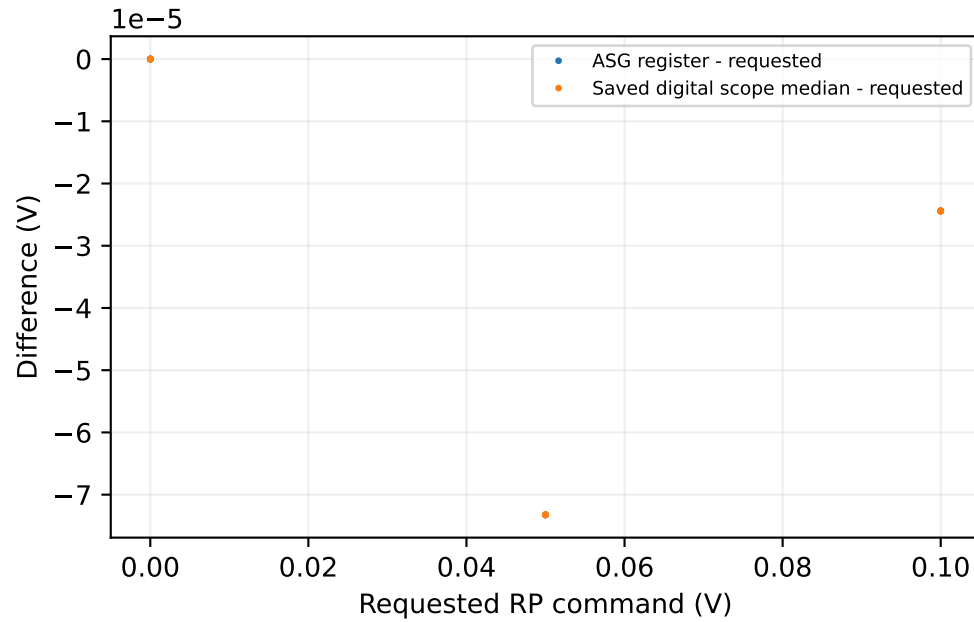
Measured phase change from each pass start



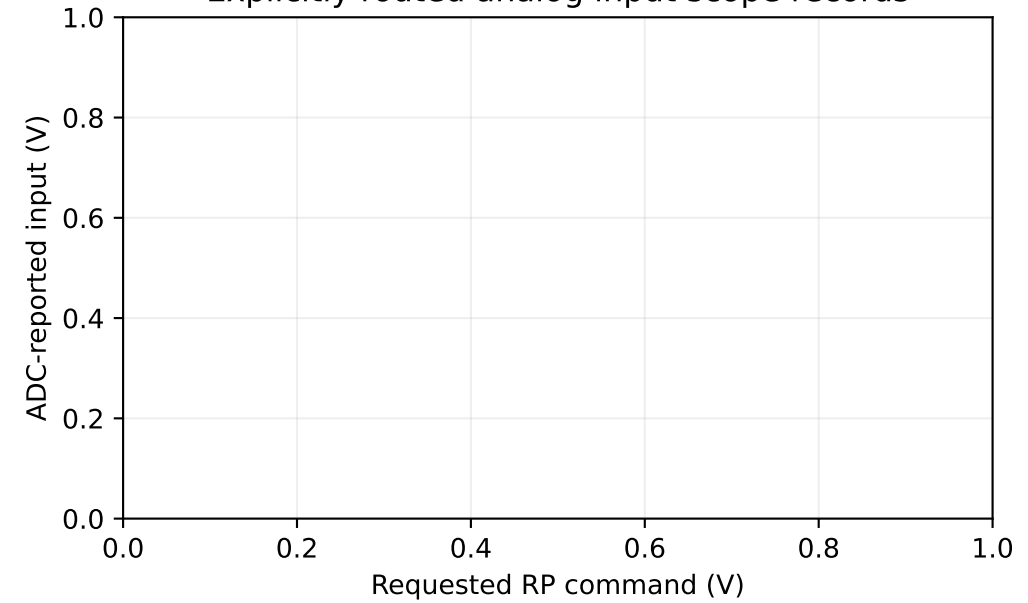
Orientation Stokes are unit-normalized from PAX angles; DOP is shown separately. No fitted or corrected trajectories.

Readbacks and direct return-path comparisons

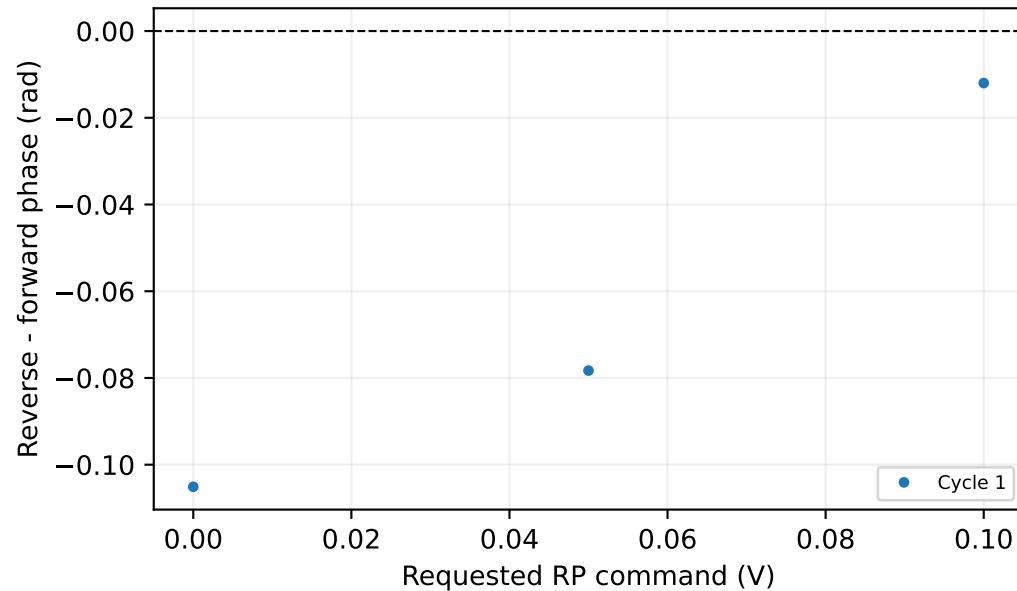
Software command versus internal readbacks



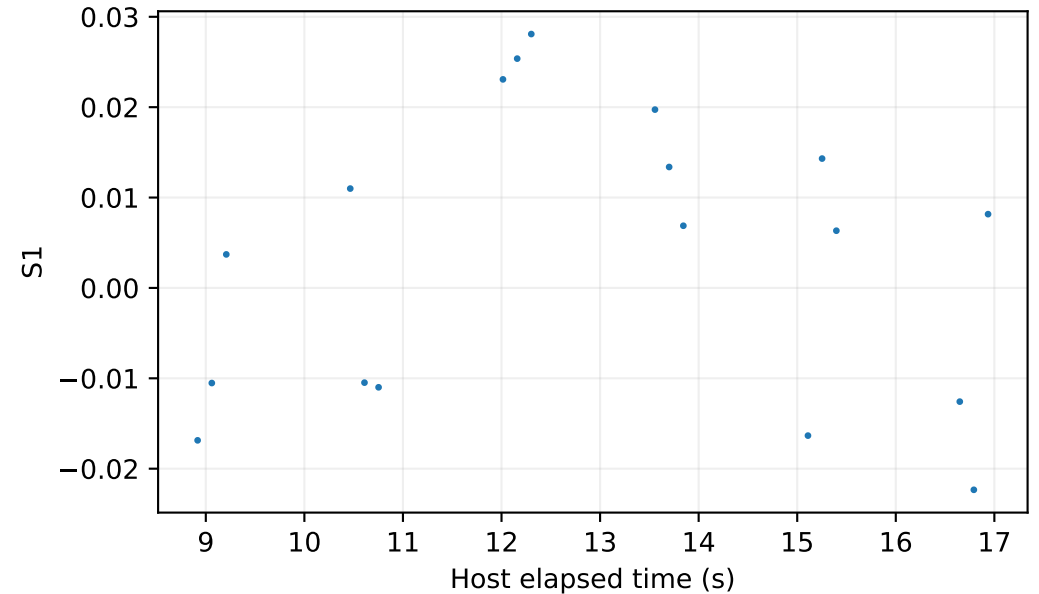
Explicitly routed analog input scope records



Same-voltage return difference: drift + hysteresis



Time record — no drift subtraction



Observed pi-crossing brackets

CYCLE 1 FORWARD — REFERENCE 0.000 V

No pi crossing bracket observed in this pass.

CYCLE 1 REVERSE — REFERENCE 0.100 V

No pi crossing bracket observed in this pass.

INTERPRETATION

Each interval brackets pi between adjacent measured step medians; no interpolation is applied. Drift, hysteresis and finite step size remain in the result. These are RP command-voltage brackets for projected equatorial phase, not calibrated terminal volts or an independent measurement of actuator retardance. Departure from $S1=0$ limits the ideal-model interpretation.

Active acquisition parameters

PROBE

False

START

0.0

STOP

0.1

STEP

0.05

REPEATS

1

SETTLE

1.0

SAMPLES

3

LABEL

readback-check

PAX_LOCATION

Physical position awaiting confirmation; no assumed port

INPUT_CONNECTIONS

IN1/IN2 physical mapping awaiting confirmation

PAX SETTINGS

Wavelength 830.0 nm; daemon mode 5; configured rotation 60.0 Hz

Active acquisition parameters

CLEANUP EVIDENCE

[0.0, 0.0]